COURSE ORIENTATION

Number Theory

CCS0047

COURSE ADVISER
3TSY2526
CCS0047 – Number Theory

Dr. Hadji J. Tejuco

TOPIC OUTLINE

- Syllabus
- Student Outcomes
- Pledge of Honor
- CS Department
- Grading System
- Assessments

- Consultation
- General Instructions on Exams
- Roll Calls
- Sync Class Schedule
- Sync Class Platforms
- Q & A / FAQs

SYLLABUS

COURSE DESCRIPTION

This course provides a comprehensive study of the fundamental properties of integers and their computational applications. It covers essential foundations ranging from the real number system and mathematical induction to advanced concepts such as divisibility theorems, modular arithmetic, and the theory of congruences. Students will explore prime distribution, primitive roots, and the law of quadratic reciprocity. The course culminates in the application of these mathematical foundations to modern cryptology, specifically the design, encryption, and decryption of secrecy systems such as the RSA cryptosystem.

COURSE LEARNING OUTCOMES

1. Prove mathematical propositions and evaluate the structural properties of integers, including divisibility, prime factorization, and algorithmic problem-solving using mathematical induction and the Euclidean algorithms.
2. Apply modular arithmetic, the theory of congruences, and advanced residue systems to solve complex systems of linear equations and computational problems.
3. Design and Evaluate secure cryptographic models, specifically public-key systems like RSA, by synthesizing number theoretic principles for data security applications.

COURSE OUTLINE

Refer to Canvas

STUDENT OUTCOMES

PROGRAM OUTCOMES

Student Outcomes

A Apply knowledge of computing appropriate to the discipline.

B Analyze a complex problem and identify and define the computing requirements appropriate to its solution.

C Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.

D Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.

E Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information Technology.

F Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.

Student Outcomes

G Analyze the local and global impact of computing information technology on individuals, organizations and society.

H Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.

I Apply knowledge through the use of current techniques and tools necessary for the IT profession.

J Apply mathematical foundations, algorithmic, principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrate comprehension of the tradeoffs involved in design choices and applicable standards.

K Apply design and development principles in the construction of software systems of varying complexity in accordance with applicable standards.

L Apply mastery on their field of specialization.

COPYRIGHT POLICIES

PLEDGE OF HONOR FOR MILES LEARNERS

- Honor Confidentiality
- Use Information Responsibly
- Protect Personal and Institutional Privacy
- Act with Integrity in Assessments
- Comply with School Guidelines

PLEDGE OF HONOR FOR MILES LEARNERS

- As a MILES learner, I recognize that I may be given access to information and resources that are confidential and protected by institutional policies. I hereby pledge to uphold the values of confidentiality, academic honesty, and responsible conduct throughout my academic journey.
- **Honor Confidentiality.** I acknowledge that all academic materials provided to me—such as laboratory activities, video lessons, presentations, and LMS content—are confidential and intended for my personal academic use only. I will take all necessary steps to protect these materials and will not share or distribute them without permission.

PLEDGE OF HONOR FOR MILES LEARNERS

Use Information Responsibly. If I am granted access to any confidential information for academic purposes, I will:

- Use it strictly for the approved purpose;
- Not copy, share, or transmit it in any format without authorization;
- Refrain from sending any materials outside the School's platforms or to unauthorized individuals.

Protect Personal and Institutional Privacy. I will maintain the confidentiality of all information obtained during classes, meetings, and discussions — including the personal information of peers and faculty. I will comply with data privacy laws and school policies at all times.

PLEDGE OF HONOR FOR MILES LEARNERS

- **Act with Integrity in Assessments.** I commit to honest conduct in all examinations. I will not seek, offer, or accept unauthorized assistance. I will follow all rules set for assessments and respect the school's policies on academic integrity.
- **Comply with School Guidelines.** I will follow all instructions given by my instructors regarding the safeguarding of confidential materials and will continue to uphold institutional rules, data privacy laws, and ethical standards.

DATA PRIVACY POLICIES

DATA PRIVACY POLICIES

- Synchronous lessons are **recorded** in MS Teams.
- Class recordings, screenshot of assessments and/or exams **ARE NOT ALLOWED** to be copied, downloaded, and distributed in any medium, even to students within the same course.
- Class recordings shall be deleted after grades encoding.
- Students **ARE NOT ALLOWED** to distribute any private and personal information that they may come across with during the conduct of the course including student and faculty information unless a direct approval has been obtained from the concerned individual.

DATA PRIVACY POLICIES

- Grades are disclosed to students only via Student Grades Portal.
- College students are mature and of legal age to decide who they want to disclose their grades (grades confidentiality).
- Faculty, Coordinators, and Directors are not obliged to answer parents' inquiries concerning grades.

COMPUTER SCIENCE DEPARTMENT

Computer Science Department

CS Director: Dr. Shaneth C. Ambat
CS Coordinator: Prof. May Florence San Pablo

Full-time Professors:

- Prof. Fanny C. Almeniana
- Prof. Anthony S. Aquino
- Dr. Angelo C. Arguson
- Ms. Kim Giselle Bautista
- Dr. Beau Gray M. Habal
- Prof. Abraham T. Magpantay
- Prof. Elisa V. Malasaga
- Ms. Hazel Patilano
- Sir Justin Jude C. Pura
- Sir Jeneffer A. Sabonsolin
- Ms. Marilyn Sanchez
- Dr. Jose Seno
- Dr. Hadji J.Tejuco

GRADING SYSTEM

GRADING SYSTEM

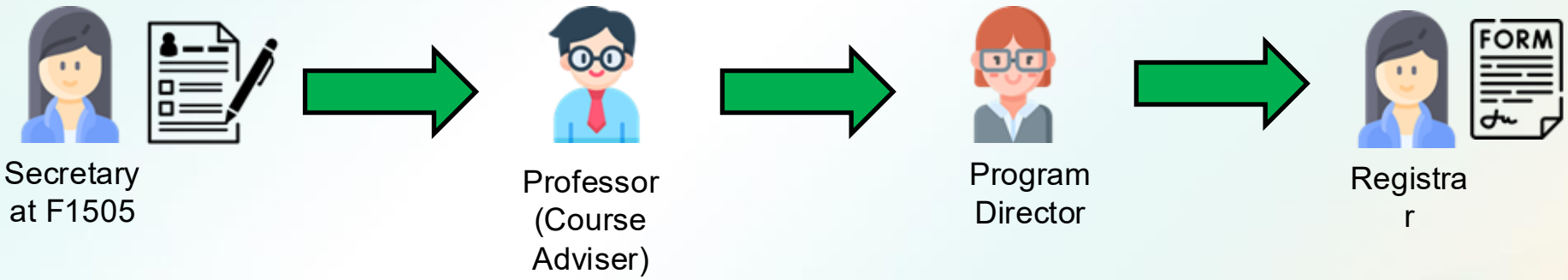
GRADING SYSTEM	
Midterm Grade	**Final Grade**
Midterm Grade (MG) = 60% (Class Standing) + 40% (Midterm Exam) **<u>Lecture: 100%</u>** **Class Standing (CS) 60%** 1. Average of at least three long quizzes 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations) **Midterm Exam (ME) 40%** **PASSING RAW SCORE: 70**	**Final Grade (FG) = 60% (Class Standing) + 15% (Midterm Exam) + 25% (Final Exam)** **<u>Lecture: 100%</u>** **Class Standing (CS) 60%** 1. Average of at least three long quizzes 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations) **Midterm Exam (ME) 15%** **Final Exam (FE) 25%** **PASSING RAW SCORE: 70**

GRADING SYSTEM

- students can check their canvas account to view their progress, find out what is missing, and even assess their grades result.
- no consultations will be done after the grade was posted on your student grades portal.

GRADING SYSTEM

- Completion of grades processing.
  - Secure and fill out a completion form at F1505 from College Secretary and have it signed by your Professor and respective Program Director before submitting it to the registrar for encoding.



GRADING SYSTEM

- Completion of grades processing.
  - Students must accomplish a completion of grades form before the final examination week of the next term.

Sample Scenario

End of First Trimester Date

Submission of Completion Form

Final Exam week of 2nd Trimester

GRADING SYSTEM

4.0	97-100	Excellent	0.5	Below 70	Failed
3.5	93-96	Superior	0.0	FDA	Excessive Absences
3.0	89-92	Very good	6.0		Passed
2.5	85-88	Good	7.0		Officially Dropped
2.0	80-84	Satisfactory	8.0		Credited
1.5	75-79	Fair	9.0		Incomplete
1.0	70-74	Barely Passing			

* FDA – Failure due to absences.

ASSESSMENT

ASSESSMENTS

- **Formative Assessment**
- Are **recorded** quiz-type of the acquired knowledge.
  Expires after 7 days of opening/prerequisite module.

- **Summative Assessment (SA)**
  A **recorded** assessment via a **quiz-type**.
  Expires after 7 days of opening (MAAS).

  * MAAS – Missed Assessment Appeal Slip

- **Project Proposal Presentation/Defense**(if applicable).

ASSESSMENTS

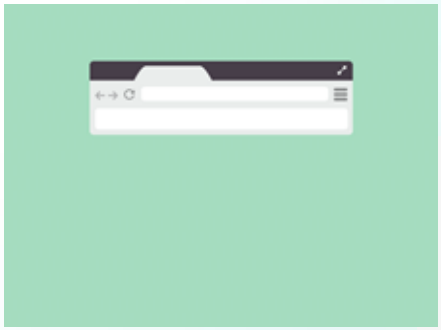
- The passing rate for **Summative Assessments**, **Midterms**, and **Final Examinations** is **70%.**

ASSESSMENTS

Summative Assessment:

- Summative assessments must be scheduled and proctored by your course adviser.
- If corrections found in SAs, kindly inform the course adviser via canvas inbox of Question Number(s) within the day.
- Only the faculty in-charge (FIC) has the right to coordinate with professors for corrections found on SAs and examinations.

ASSESSMENTS

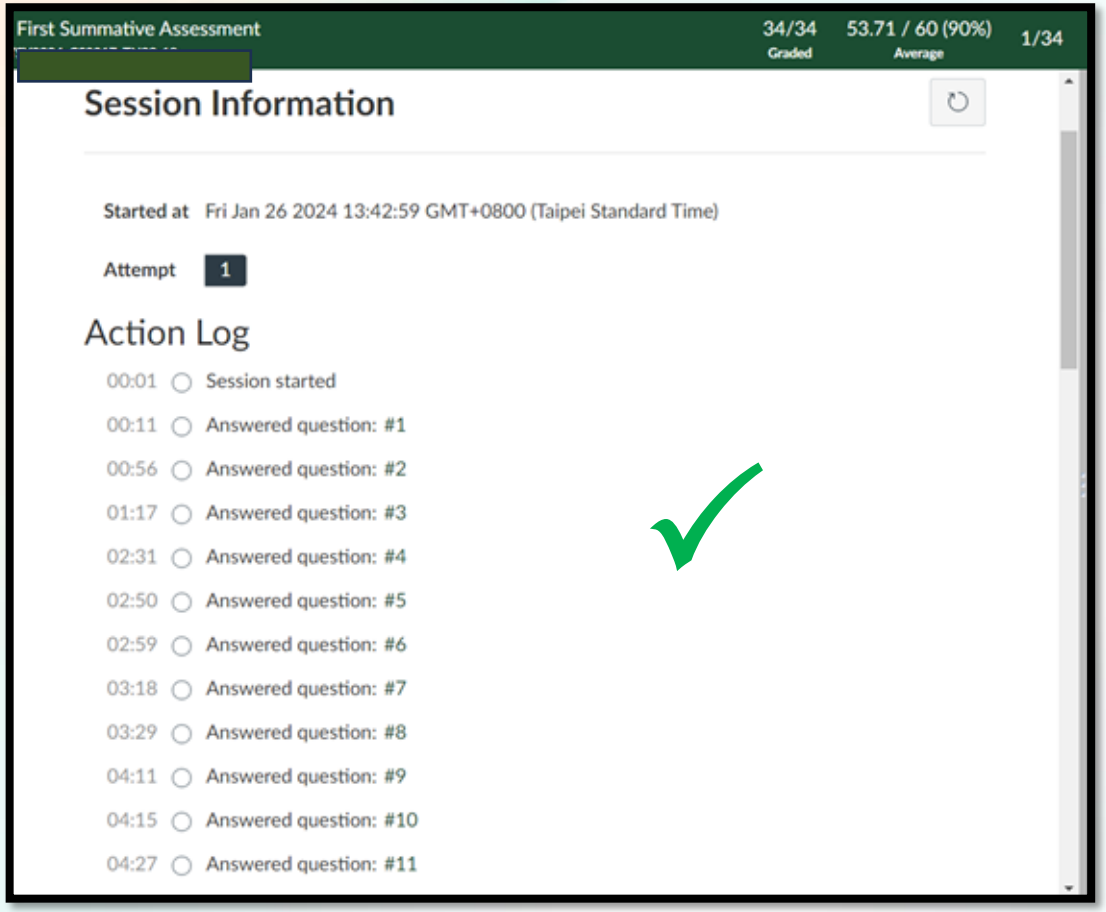


Summative Assessment:

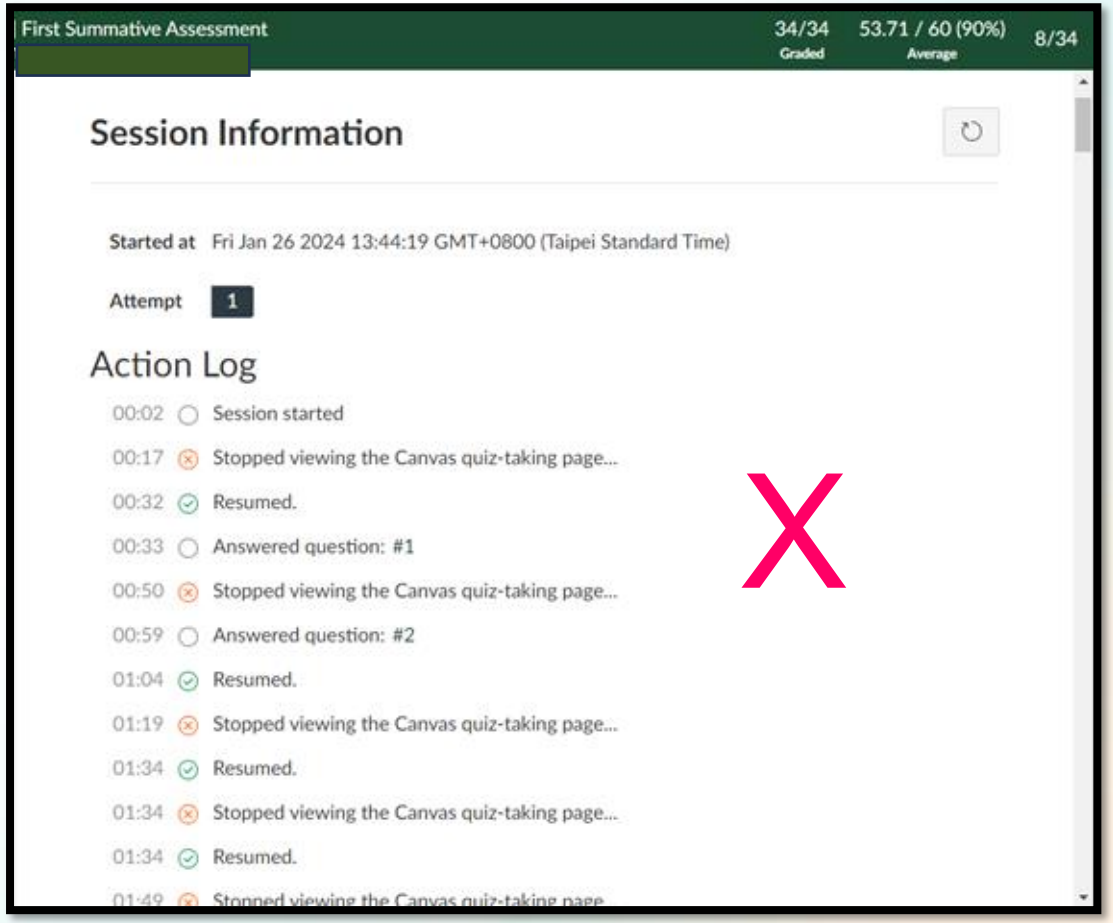
- Place all things (gadgets) inside your bag.
- Maintain one browser with one tab only with no running applications, pop-ups, and sidebar search engine (i.e. copilot).
- Refrain from using AI-tools or equivalent.

- Any suspicious activities are flagged for evaluation.
- Receiving a red flag is automatic **0** point in that assessment.
- Maintain a clean log at speed grader (see next slide).

ASSESSMENTS



Good. No signs of switching tabs or apps while taking an assessment.



Cheating. Noticeably cheating by switching app views while taking an assessment.

ASSESSMENTS

Summative Assessment (Online):

- Make sure you have a functioning webcam and microphone.
- Turn off all other devices and place them outside of your reach.
- Clear your workspace of all other materials.
- Make sure you are in a well-lit room.
- Your eyes should stay on the screen at all times.

ASSESSMENTS

Summative Assessment (Online):

- Close all applications except for Canvas and MS Teams application.
- Open your webcam all through out the duration of the Summative Assessment.
- Ensure that your face is clearly visible on the web camera while positioning it directly facing you.

ASSESSMENTS

Summative Assessment (Online):

- Make sure that your web camera shows your entire face down to your collarbone.
- Using a smartphone, tablet, laptop, or any gadget as alternative web camera is prohibited.
- Do not wear sunglasses or hats or anything that will conceal your identity.

ROLL CALLS

Roll Calls

Done using OCRS grading system portal.

Conditions for attendance:

- Must be responsive during roll calls.
- Inform your Prof via inbox **per instance** of being absent.
  - Excuse letter with signature and proofs needed.
- Allowed absences with only **20%** of total attendance.
  - Attendance dictates the final remarks (INC, DROPPED, FDA)

Roll Calls

Allowed (80% Present)
21 days = 17 days
22 days = 18 days
23 days = 18 days
24 days = 19 days
25 days = 20 days
26 days = 21 days
27 days = 22 days
28 days = 22 days

Note: Late & Absent are not counted in the system.

2 lates = 1 absent.

Note: 14 weeks of classes are reflected on school calendar.

Roll Calls

- If marked absent, it is the student's **responsibility** to submit a School Physician-verified excused letter to their course adviser within a week of absence.

- Late enrollees must provide proof (i.e. time stamp on registration form/canvas/email conversation with Registrar staff) to their course adviser for being late enrolled in the course.

GENERAL INSTRUCTIONS ON EXAMS

GENERAL INSTRUCTIONS ON EXAMS

- Check your Final Examination schedule and rooms at **Registrar's canvas portal** or **Student Plaza postings**. The Professor may post an examination schedule via Canvas Announcement.

- Taking the examination in advance due to whatever reason is discouraged and is strictly prohibited since it can lead to leakage of the said examination.

GENERAL INSTRUCTIONS ON EXAMS

- Conflict in examinations must be settled with their respective Professor to ensure same date was followed and it does not exceed to the Professor's logout time.

Professo
r

- For invalid reason, the student will be advised to take it after the current term. As a result, the student is marked as "Incomplete" automatically for the current term.

GENERAL INSTRUCTIONS ON EXAMS

- A Proctor has the right to report any incident found during examinations and cannot be contested. No warning is needed since cheating is an act of grave offense.

Professo
r

- Strictly implemented: Any means of cheating (looking, talking, gadgets, browsing, etc.) is prohibited.

GENERAL INSTRUCTIONS ON EXAMS

- Use the restroom before the exam begins. Leaving the exam room for washroom break is highly discouraged.

- Student caught red handed will be sanctioned according to student handbook Section 7.4 Penalties which will result to expulsion.

CONSULTATION

CONSULTATION

- Adhere to Professor's Consultation Hours (to be posted in our Canvas soon).

- Consultations will be conducted Face-to-Face at F1505.

2TSY2223_CCS0005_TN02-2

FACULTY CONSULTATION HOURS	
Monday	
Tuesday	08:00 - 09:40
Wednesday	
Thursday	
Friday	
Saturday	

Sample Only

* Consultation via iCare is allowed for Prof with less than 40 hours of teaching within a week. Ask your Prof for possible work schedule arrangement.

Missed Assessment

Step 1
Submit your medical proofs to FEU Clinic for verification.

1

Note:

- Must be taken within 7 days after the scheduled SA.

  Valid reason: Health-related issues.

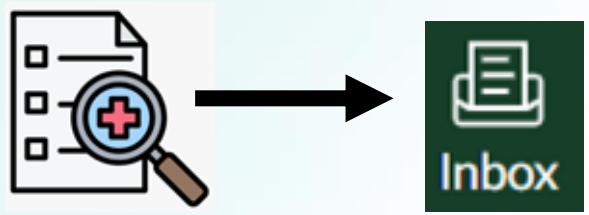
Missed Assessment

Step 2

Send a formal request via Canvas Inbox.

Attachments:

- Accomplished MAAS (see canvas for template).
- FEU Clinic-verified medical proofs.



1 **2**

* FEU Clinic a.k.a. Health and Services Unit (HSU)

Missed Assessment

Step 3
Wait for your approval via Canvas inbox.

1

2

3

Approval Guide:

- 1st Attempt: Course Adviser approval
- 2nd Attempt: Program Director approval
- 3rd Attempt: Senior Director approval

Missed Assessment

Step 4

Take your Special Assessment F2F.

- Your Prof will inform you about the schedule at iCare.
- Check your Prof consultation schedule (see Canvas).

1 2 3 4

SYNC CLASS SCHEDULE

SYNC CLASS PLATFORM

SYNC CLASS PLATFORM

- MS Teams link will be posted via Announcement in Canvas
- Use your IDNumber@fit.edu.ph email account when joining.
- Guest accounts will **NOT** automatically be admitted.

Q&A SESSION

Frequently Asked Questions:

Q: What if a student got sick/accident during Final Examination day?

A: Students with health-related reason(s) are advised to prioritized recovery. They will be advised to be receive INC grade then complete it on the next term.

Q: Do I need to retake a course if I received 9.0 (INCOMPLETE)?

A: Yes and No.

Yes, if having complied everything and still failed.

No, if able to receive a passing grade after complying.

Frequently Asked Questions:

Q: Will my MAAS be approved if I have medical certificate?

A: No, medical certificate must be verified **FIRST** by the HSU.

Q: What if I am fully recovered, should I go now for F2F?

A: Not yet, Only if you have the School Clinic approval.

Frequently Asked Questions:

Q: What if my roll calls was below 80% even before Midterm Exam?

A: Students are advised to submit a printed letter of consideration to course adviser signed both student and guardian with legal ID to be accepted on the succeeding classes.

Q: What to expect if my Undertaking was INC/FAILED even on Midterm Exam week?

A: Submit a printed letter for consideration signed both student and guardian with legal ID.

Frequently Asked Questions:

Q: I did not submit medical proof to school clinic because I don't want anyone to find out my health-related reason for being absent?

A: Students are required to submit medical proofs as basis for attendance and will help the school aiding them in the future in case of emergency. Medical proofs are returned to students after verification process. With this, health-related reasons are only disclosed to School Clinic and Professor.

thank
you!

Next meeting:

- Prepare for your Class Introduction.
- Deadline for updating your formal profile photos in MS Teams and Canvas.